constant observation, undermining the very social and ethical foundations that technology aims to strengthen.

7.5.2 Legal responsibility and liability in autonomous drone operations

The growing autonomy of drones, driven by advancements in artificial intelligence, machine learning, and onboard decision-making, has introduced complex questions about legal responsibility and liability. Unlike conventional drones that require human pilots, autonomous drones can make real-time decisions with minimal human oversight, raising uncertainties about who should be held accountable when accidents, damages, or unlawful actions occur. This issue lies at the intersection of technology, law, and ethics, demanding the evolution of legal frameworks capable of addressing the unique nature of AI-driven systems.

Traditionally, responsibility for UAV-related incidents rested with the human operator or pilot, who was directly in control of the aircraft's actions. However, in the case of autonomous drones, the decision-making process is often delegated to algorithmic systems that interpret data, plan routes, and respond to environmental conditions independently. When a drone malfunctions, violates airspace regulations, or causes harm, it becomes difficult to determine whether liability lies with the operator, manufacturer, software developer, data provider, or AI algorithm itself. This phenomenon, often termed the "accountability gap," poses significant challenges to policymakers and insurers.

Legal systems around the world are beginning to address this ambiguity by exploring shared or distributed liability models. Under such frameworks, responsibility is proportionally assigned based on the source of failure:

- Manufacturers may be liable for hardware defects or design flaws.